TECH REPORT

ECG-Only Brugada Syndrome Screening on Brugada-HUCA
Using Transfer Learning and
Safety-Oriented Model Selection Framework

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Abstract

This project benchmarks ECG-driven machine learning and deep learning models for binary Brugada syndrome screening on the Brugada-HUCA dataset. We deliberately use an ECG-only primary benchmark to avoid shortcut learning from metadata such as `basal_pattern` and `sudden_death`. After excluding seven label-2 atypical cases, the final binary benchmark contains 356 patients (287 Normal, 69 Brugada), evaluated with one shared patient-level split across 11 models. In the final rerun, an in-domain transfer-learning pipeline emerged as the strongest aggregate, clinical-safety, and balanced multi-aspect candidate (PR-AUC 0.676, F1 0.692, sensitivity 0.643, specificity 0.948). Feature-model interpretability still emphasizes V2 and V1, while sequence saliency for the final transfer-learning classifier is more distributed, with V1 strongest and additional signal on V2 plus selected limb leads. We therefore position the system as research-stage decision support rather than autonomous diagnosis.

1 Problem Statement and Dataset Justification

The task is to distinguish Brugada syndrome from Normal 12-lead ECG recordings while balancing two competing goals: strong aggregate discrimination and low false-negative risk. In this setting, missing a true Brugada case is more clinically costly than generating an additional false alarm, so leaderboard performance alone is not sufficient.

Brugada-HUCA is appropriate for this challenge because it directly provides short 12-lead ECGs for a Brugada-focused cohort in WFDB format, enabling end-to-end signal modeling, feature engineering, and clinically grounded interpretability. The repository-level checks confirmed that the recordings are standardized at 100 Hz, 12 seconds, and 12 leads for all indexed subjects. The metadata file contains 363 subjects in total: 287 Normal, 69 confirmed Brugada (`brugada`=1), and 7 atypical/other cases (`brugada`=2). Because the competition problem is binary Brugada-vs-Normal classification, we excluded the seven label-2 cases and evaluated the models on a clean 356-patient binary cohort.

Table 1: Benchmark cohort and split summary.

Item	Value
Subjects in metadata	363
Excluded atypical label-2 cases	7
Binary benchmark subjects	356
Normal / Brugada	287 / 69
Train / Validation / Test	227 / 57 / 72
Brugada cases in test split	14
ECG-only feature count	357
Median-beat tensor shape	70×12

The main comparison excludes metadata such as `basal_pattern` and `sudden_death` so that the reported benchmark remains ECG-driven, fair, and clinically defensible. Metadata is discussed only as a supplementary ablation because it can improve apparent performance without proving better ECG understanding.

2 Methodology

Our pipeline follows one shared patient-level stratified split for every model so that comparisons remain fair. We first verified ECG availability and recording specifications for all patients, then loaded each record from WFDB and standardized the lead order to the conventional 12-lead layout.

Preprocessing. Each ECG was filtered with a third-order zero-phase Butterworth band-pass filter from 0.5–40 Hz, then normalized with robust median/MAD scaling. R-peaks were detected from a priority list of reference leads (II, V2, V1, V3, I) using a smoothed signal plus prominence-aware peak finding. Around each accepted beat, we extracted a window of 0.25 s before and 0.45 s after the R-peak, and the patient representation was formed as the median beat across detected beats. This keeps the core morphology while suppressing transient noise.

Feature representations. We built two ECG representations. The first is a 357-dimensional ECG-only feature table that combines per-lead strip statistics, beat morphology descriptors, right-precordial ST/QRS features, and selected cross-lead correlations. The second is a 12-lead median-beat tensor of shape 70×12 used by end-to-end sequence models.

Table 2: Held-out test comparison between the best aggregate model and the best safety-oriented clinical candidate.

Track	Model	Thr.	PR-AUC	F1	Sens.	Spec.	FN	FP
Aggregate	Transfer Learning (DAE)	0.354	0.676	0.692	0.643	0.948	5	3
Clinical	Transfer Learning (DAE)	0.354	0.676	0.692	0.643	0.948	5	3

Model families. We trained 11 ECG-only models in three families: feature-based models (XGBoost, Random Forest, SVM-RBF, AdaBoost), end-to-end deep models (1D CNN, ResNet-1D, CNN+BiGRU, Transformer Encoder), and advanced/transfer approaches (in-domain denoising-autoencoder transfer learning, VICReg self-supervision, and Echo State Network).

Imbalance handling and thresholding. Because Brugada is the minority class, we used stratified splitting, class-balanced training weights, `scale_pos_weight` for XGBoost, weighted AdaBoost fitting, and evaluation metrics that do not rely on accuracy alone. Probabilistic thresholds were tuned on the validation split, not the test split, by maximizing F1 from the precision-recall operating points. The fixed validation-derived threshold was then applied once to the held-out test set. This is stricter than post hoc test-set threshold selection.

Evaluation strategy. We report both a leaderboard-oriented ranking and a safety-oriented clinical ranking. The first emphasizes PR-AUC, F1, sensitivity, balanced accuracy, and correct Brugada detections. The second prioritizes sensitivity and false-negative control. Because multiple models were compared on a single held-out test split, the leaderboard should be interpreted as a useful benchmark rather than a definitive estimate of out-of-sample superiority.

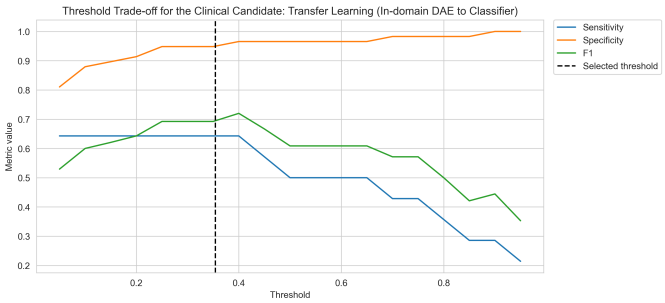
For reproducibility, the generated benchmark artifacts are organized in the project repository [1] under `outputs/` (leaderboards and logs), `outputs/predict` (predictions and detailed analysis tables), and `outputs/plots` (figures used in this report).

3 Results and Validation

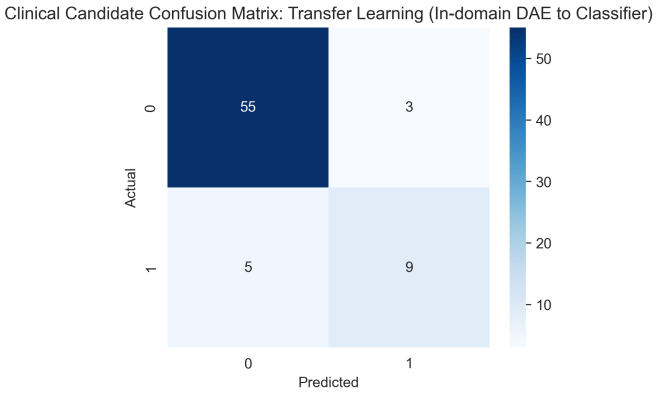
All 11 candidate models trained successfully on the final pipeline. In this rerun, the aggregate winner and the safety-oriented winner converged on the same model, which is helpful rather than trivial: it means the top overall model also provided the strongest false-negative profile among the tested candidates on this split.

The transfer-learning model now leads the aggregate leaderboard because it combines the strongest precision-recall behavior on this split with a clinically competitive operating point: it detects 9 of 14 Brugada cases while keeping false positives to 3. Unlike the earlier run, no separate clinical winner is needed to tell the screening story because the same model already gives the best balance between missed cases and unnecessary alarms among the tested finalists.

A balanced multi-aspect ranking that blends aggregate performance, clinical safety, validation consistency, training effort, and available repeated-split evidence also selected the transfer-learning pipeline, with a composite score of 0.720. This means the same transfer-learning model is now the aggregate winner, the clinical candidate, and the final single-model recommendation.



(a) Clinical threshold trade-off for the transfer-learning candidate.



(b) Confusion matrix at the validation-derived threshold.

Figure 1: Held-out test analysis for the safety-oriented clinical candidate.

Threshold rationale. The safety-oriented transfer-learning threshold was selected on the validation split at 0.354. On the held-out test set, that operating point detected 9 of 14 Brugada cases (sensitivity 0.643) and correctly rejected 55 of 58 Normal cases (specificity 0.948). A descriptive threshold scan on the same test predictions shows that nearby thresholds around 0.40 can preserve sensitivity while slightly improving specificity on the test split, but those analyses are presented only to visualize the trade-off and not to retroactively claim a better final threshold.

Robustness beyond a single split. To assess stability on a small and imbalanced cohort, we supplemented the shared hold-out benchmark with six repeated patient-level stratified outer splits on the train+validation pool. To keep runtime manageable, this stability check focused on fast ECG-only finalists. The repeated-split summary in Table 3 shows that ESN delivered the highest mean sensitivity, whereas XGBoost achieved the strongest mean PR-AUC and the lowest false-negative variance among the tested lightweight models.

This added stability analysis improves rigor, but it does not fully eliminate the model-selection optimism that can arise when many candidate models are compared against one final hold-out split.

Table 3: Repeated split stability on the train+validation pool (mean ± std).

Model	PR-AUC	Sensitivity	False Negatives
ESN	0.535 ± 0.072	0.619 ± 0.195	5.33 ± 2.73
XGBoost	0.560 ± 0.105	0.512 ± 0.114	6.83 ± 1.60
Random Forest	0.519 ± 0.119	0.452 ± 0.224	7.67 ± 3.14

Metadata ablation. Adding shortcut-risk metadata produced inconsistent effects: for example, XGBoost sensitivity fell from 0.571 to 0.357, while SVM sensitivity rose from 0.214 to 0.500 with worse specificity. Because these gains are not stable and may reflect shortcut learning rather than stronger ECG reasoning, the main benchmark remains ECG-only.

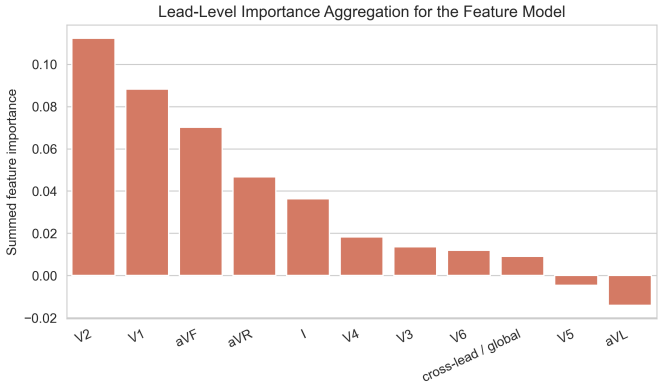
4 Interpretability and Clinical Impact

Interpretability was designed to answer a clinically relevant question: are the models focusing on plausible Brugada regions rather than arbitrary patterns? For the best interpretable feature model (XGBoost), permutation importance aggregated by lead ranked V2 first, followed by V1, with additional contribution from aVF and aVR. Within the right precordial feature set, important variables included `V2_minus_V3_st_mean`, `V1_strip_kurtosis`, `V2_strip_max`, and several V1 morphology descriptors. This remains compatible with the expectation that Brugada-related morphology is often visible in V1–V3.

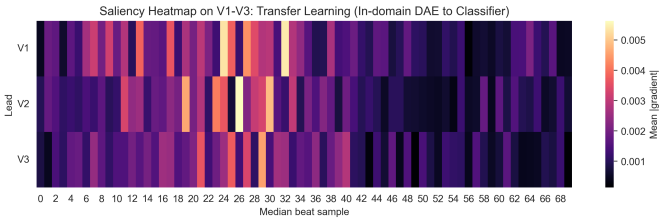
For the final transfer-learning sequence model, gradient-based lead saliency over the top Brugada-positive test predictions ranked V1 first, followed by aVF, aVL, and V2, with V3 still present but not dominant. This is more distributed than the previous ResNet-centered story, so the interpretability conclusion should be more cautious: the model still pays attention to clinically relevant precordial structure, but it also uses broader cross-lead information rather than concentrating almost exclusively on V1–V3.

These interpretability outputs should be read as descriptive transparency tools. They help assess whether the model is focusing on clinically plausible ECG regions, but they do not prove causal reasoning or guarantee safe deployment.

From a clinical workflow perspective, the transfer-learning candidate is now also the strongest screening story on this split because it catches 9 of 14 Brugada cases while adding only 3 false positives. However, it still leaves 5 false-negative Brugada cases. Therefore, the current model is better framed as a triage aid or second-reader signal for specialist review, not as an autonomous diagnostic system.



(a) Lead-level permutation importance from the XGBoost feature model.



(b) Transfer-learning saliency heatmap on the inspected V1–V3 leads.

Figure 2: Interpretability outputs retain right-precordial relevance while showing broader cross-lead use in the final sequence model.

5 Limitations and Responsible Deployment

This study has several important limitations. First, the cohort is small and imbalanced, and the benchmark is based on a single dataset without external validation. Second, the binary setup excludes seven atypical label-2 subjects, which simplifies the real diagnostic spectrum. Third, repeated split analysis was limited to lightweight finalists for runtime reasons, so deep-model stability remains an open question. Fourth, many candidate models were compared on one held-out test split, which introduces selection optimism even when the split itself is untouched during threshold tuning.

For responsible deployment, any future system should undergo external validation, prospective operating-point calibration, subgroup analysis, and workflow design that clearly assigns final authority to clinicians. In short, the present results show promise for interpretable ECG-based decision support, but they are not sufficient for standalone diagnosis.

References

[1] A. Nashrudin and R. Muslimah, "IDSC2026," GitHub repository, 2026. Available: https://github.com/avezoor/IDSC2026.

[2] N. Costa Cortez and D. Garcia Iglesias, "Brugada-HUCA: 12-Lead ECG Recordings for the Study of Brugada Syndrome," version 1.0.0, PhysioNet, 2026, RRID:SCR_007345. doi: 10.13026/0m2w-dy83.

[3] A. Goldberger, L. Amaral, L. Glass, J. Hausdorff, P. C. Ivanov, R. Mark, et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000, RRID:SCR_007345. doi: 10.1161/01.CIR.101.23.e215.